

Aditya Vasu Rana

Chennai, India | adityavasurana@gmail.com | 7876007402 | linkedin.com/in/aditya-rana | github.com/adityaa-rana

Education

Vellore Institute of Technology, Chennai

Aug 2023 – May 2027

Bachelor of Technology in Computer Science with specialization in Data Science, **CGPA : 9.45 (Branch Rank 3)**

- **Coursework:** Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Machine Learning, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence

Technical Skills

- **Programming Languages:** Python, R, C, C++ ,Java, SQL
- **Core AI/ML:** Machine Learning, Neural Networks, Transformers, Deep Learning, NLP, Computer Vision
- **Tools & Platforms:** Linux, Git, AWS, Docker, Jupyter, Streamlit , Hex
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, HuggingFace
- **Databases & Visualization:** PostgreSQL, PL/SQL, Tableau, Power BI, Data Storytelling
- **Web Technologies:** HTML, CSS, JavaScript, ReactJS, Flask, FastAPI, Tailwind, Bootstrap

Projects

AI Research Paper Understanding Platform - RAG + Transformers + FastAPI [↗](#)

- Developed an end-to-end Research Paper Intelligence platform using **RAG**, integrating **MiniLM embeddings**, **FAISS**, and **FLAN-T5** for grounded summarization and question answering.
- Built a document processing pipeline with **PDF parsing**, chunking, semantic embedding generation, and vector indexing for efficient context-aware retrieval.
- Implemented AI services for semantic search, summarization, quiz generation, keyword extraction, and equation/figure explanation using **FastAPI** and **PostgreSQL**.

AI-Powered Applicant Tracking System (ATS) - FastAPI + React.js + PostgreSQL + NLP [↗](#)

- Built a **full-stack recruitment platform** with **FastAPI**, **React.js**, **PostgreSQL**, **SQLAlchemy**, and **JWT authentication**, enabling role-based workflows for recruiters and candidates.
- Developed an automated **resume-job matching pipeline** using **PyMuPDF** and NLP techniques to extract skills, generate **ATS scores**, and rank candidates based on job description relevance.
- Implemented application tracking, candidate ranking, shortlisting/rejection workflows, recruiter feedback, and interview question generation to streamline the hiring process.

Fashion Recommender System - CNN + ResNet50 + KNN + FastAPI Backend [↗](#)

- Built a **content-based fashion recommendation system** using **ResNet50 (ImageNet pretrained)** as a feature extractor to generate **2048-D deep embeddings** for 44000+ fashion product images.
- Implemented an efficient image retrieval pipeline using **GlobalMaxPooling**, **L2 normalization**, and **K-Nearest Neighbors (KNN)** to recommend the **Top-5 visually similar items**.
- Developed an interactive **Python FastAPI web application** that allows users to upload a fashion image and receive real-time recommendations using precomputed embeddings stored in `embeddings.pkl`.

Experience

Data Science and Analytics (D2C Insights) Intern, Kerala Ayurveda Ltd.

Oct 2025 – Jan 2026

- Analyzed D2C sales funnel across multiple product categories using Python and SQL, identifying key drop-offs such as low Add-to-Cart (ATC) and Cart-to-Checkout (C2C) rates for sales improvement initiatives.
- Conducted customer behavior analysis including low ATC diagnosis, abandoned cart AOV analysis, and first-product to next-likely-product mapping, generating actionable insights to optimize conversion journeys.
- Collaborated closely with the product management team to frame hypotheses, validate marketing strategies, and support data-driven decisions aimed at improving sales performance and customer engagement.

Research Experience – Mango Variety Classification using Swin Transformer

2026

- Conducted research on fine-grained mango variety classification using a Swin Transformer, integrating preprocessing, data augmentation, weighted loss, and transfer learning.
- Achieved 98.04% test accuracy across 15 classes with comprehensive evaluation using precision, recall, F1-score, and confusion matrix, documenting the methodology and experimental analysis.

Certifications and Achievements

- Frontier Research Paper on Mango Variety Classification using Vision Transformer-based architectures.
- 2nd place at V-Medithon Intercollege Hackathon, Ranked 1 for Machine Learning model.
- 300+ problems solved on LeetCode with two consecutive 100 Days Badges (2025–2026). [↗](#)
- **Oracle Cloud Infrastructure Architecture: Data Science Professional**, Oracle University [↗](#)
- Patent accepted for publication on an IoT sensor-based smart biodegradable milk packaging system.